MEDILINK HEALTH NETWORK

ENTERPRISE DATA GOVERNANCE FRAMEWORK

Project: MediLink Health Network Enterprise Data Platform
Document Type: Data Governance Framework
Status: Proposed Framework for the Project Case Study

1. Governance Charter

1.1 Purpose

The MediLink Health Network Data Governance Framework establishes the principles, responsibilities, policies and procedures required to manage organizational data throughout its lifecycle. It provides a structured approach to ensuring that data is accurate, consistent, secure, appropriately accessible and suitable for operational and management decision-making.

The framework supports the integration of patient, clinical, billing and operational data from multiple healthcare facilities into a centralized enterprise data platform. It establishes accountability for data ownership, defines standards for patient identity management, and provides mechanisms for monitoring data quality, protecting sensitive information and managing compliance risks.

The framework also establishes governance requirements for the movement of data from source systems through staging and master data management into the enterprise data warehouse and reporting environment.

1.2 Governance Objectives

The framework has the following objectives:

1. Establish clear ownership and accountability for organizational data.
2. Improve patient identity consistency through master data management and controlled duplicate resolution.
3. Define common data definitions, formats, validation rules and quality standards across participating facilities.
4. Protect confidential and personally identifiable information through appropriate access controls and security safeguards.
5. Establish documented policies for data access, retention, archival, disposal and incident management.
6. Ensure that data is documented, validated and assessed before being accepted as trusted enterprise information.
7. Establish procedures for identifying, recording, escalating and resolving data-quality and compliance issues.

8. Improve the reliability, consistency and timeliness of consolidated management reports.
9. Support regulatory compliance, auditability and evidence-based decision-making.

1.3 Scope

The framework covers the following areas:

- **Patient master data:** Patient identity, demographic information, contact details and source-system identifiers.
- **Clinical data:** Patient visits, diagnoses and related clinical information.
- **Billing data:** Healthcare services, payment modes and recorded financial transactions.
- **Source systems:** Data originating from participating healthcare facilities.
- **Data integration:** Extraction, staging, standardization, validation and loading processes.
- **Master data management:** Patient matching, golden-record creation, source-record linkage and stewardship review.
- **Enterprise data warehouse:** Dimension tables, fact tables and analytical data structures.
- **Metadata and lineage:** Data definitions, ownership, transformation rules and traceability.
- **Business intelligence:** Dashboards, management reports and data-quality monitoring.
- **Security and privacy:** Access management, protection of sensitive information, audit logging and data lifecycle management.

For the simulated Mulago case study, this framework represents a proposed governance design. It does not claim that Mulago Hospital or any other real healthcare organization has adopted these arrangements.

1.4 Governance Principles

MediLink's proposed governance arrangements are guided by six principles.

Principle	Application
Accountability	Every critical dataset has an assigned owner responsible for its appropriate use and quality.
Data quality	Data is assessed against defined completeness, validity, consistency and uniqueness requirements.
Privacy and confidentiality	Personal and clinical information is accessed and processed only for authorized purposes.
Standardization	Common definitions, formats and validation rules are applied across participating facilities.
Traceability	Data sources, transformations, corrections and significant access events are documented where appropriate.
Continuous improvement	Quality results, audit findings and unresolved issues inform corrective action and periodic review.

These principles are consistent with the accountability, quality and security requirements reflected in Uganda's data protection legislation.

2. Governance Structure and Responsibilities

The proposed governance structure separates strategic accountability, business ownership, data stewardship and technical administration. This separation helps prevent data-quality and compliance responsibilities from being concentrated exclusively within the IT function.

2.1 Governance Roles

Role	Main responsibilities
Chief Data Officer or designated Data Governance Lead	Coordinates the governance programme, oversees policy implementation, reviews significant governance issues and reports material risks to management.
Data Owner	Is accountable for the appropriate use, quality, classification and access requirements of data within an assigned business domain.
Data Steward	Maintains data definitions and quality rules, investigates data-quality exceptions, coordinates corrections and manages referrals involving conflicting patient records.
IT Custodian	Maintains the technical infrastructure, manages database permissions, implements approved security controls, supports backups and maintains relevant technical logs.
Clinical Representative	Ensures that clinical data definitions and corrections are appropriate for clinical use and reports errors that could affect patient care.
Facility Data Representative	Coordinates data-quality activities at an individual facility, investigates source-data errors and communicates unresolved issues to the appropriate steward.
Report or Analytics Owner	Validates report definitions, verifies calculations against approved data sources and documents known reporting limitations.
Privacy or Compliance Lead	Coordinates privacy-risk assessments, monitors compliance obligations and supports incident handling, subject to the organization's formal allocation of responsibilities.

These roles are proposed for the project. They should be assigned to actual personnel, with appropriate authority and documented responsibilities, before implementation in a real organization.

2.2 Decision-Making and Escalation

Routine data-quality issues should initially be handled by the relevant data steward or facility representative. Issues requiring changes to approved data definitions, master records, access permissions or business rules should be escalated to the responsible data owner.

Potential privacy incidents, unauthorized disclosures, significant security weaknesses and unresolved compliance risks should be escalated promptly to the designated security and privacy personnel and the appropriate management authority.

The Data Governance Lead should coordinate significant cross-facility issues and ensure that decisions are documented. Decisions affecting clinical interpretation or patient identity should involve the relevant clinical or data owner rather than being made solely by technical personnel.

A governance issue should remain open until its corrective action has been completed, its outcome has been checked and the appropriate person has approved closure.

2.3 Governance Records

The proposed governance process should maintain the following records:

- Data ownership and responsibility register.
- Business glossary and data dictionary.
- Data-quality rules and monitoring results.
- Master data management exception and resolution records.
- Access approvals and periodic access reviews.
- Compliance and risk register.
- Incident records and corrective-action reports.
- Data retention and disposal records.
- Change approvals and report-validation evidence.

These records provide evidence that governance decisions are traceable and that identified issues are being managed systematically.

3. Governance Policies and Control Requirements

The framework establishes the following core policies. Each policy should be supported by documented procedures, technical controls and monitoring evidence.

3.1 Patient Identity and Master Data Management Policy

Purpose: To maintain consistent patient identities across participating facilities while preventing incorrect record merging and preserving source-system traceability.

The policy requires the following:

1. Each accepted patient master record must have a unique internal patient key.
2. Patient records from different source systems must be matched using approved identity-matching rules.
3. Standardization should be applied to relevant fields, including telephone numbers, names and other approved matching attributes.
4. Records identified as possible duplicates should be assessed using defined matching criteria.
5. High-confidence matches may be processed according to approved automated matching rules, while ambiguous or conflicting matches must be referred to a data steward.
6. Source-system identifiers and record-linkage information must be retained to support traceability.
7. Changes to patient identity information must be recorded where required by the audit policy.
8. A record must not be merged solely because one matching attribute is identical. The matching process should consider the risk of incorrectly combining different patients.

For the implemented project schema, the `m_patient` table represents the proposed golden patient record, while `patient_source_links` maintains links between the master record and its source records. The `mdm_review_queue` supports the review of unresolved matching cases.

Control evidence: Patient-matching rules, duplicate-review records, source-linkage checks and test results demonstrating that ambiguous matches are referred for review.

3.2 Data Quality and Standardization Policy

Purpose: To establish consistent requirements for assessing and improving the quality of enterprise data.

The policy requires data to be assessed against the following dimensions:

- **Completeness:** Required fields contain values where applicable.
- **Validity:** Values conform to approved formats, ranges and permitted categories.
- **Consistency:** Equivalent information is represented consistently across systems.
- **Uniqueness:** Records that represent the same real-world entity are identified and appropriately resolved.
- **Accuracy:** Information is checked against reliable sources or approved verification procedures where possible.
- **Timeliness:** Data is available within the required operational and reporting periods.

Data-quality rules should be defined for critical attributes before the data is accepted for trusted reporting. Exceptions should be recorded and assigned to the appropriate data steward or data owner.

For example, telephone numbers should be standardized to the agreed Ugandan international format where the underlying number can be reliably established. Invalid or ambiguous numbers should be flagged for review rather than silently modified.

The project may use proposed targets such as telephone-number completeness of at least 97%. Such targets must be distinguished from measured results, and actual performance should be reported using the results produced by the implemented database.

Control evidence: Data-quality rules, profiling results, scorecard outputs, exception records and before-and-after validation results.

3.3 Access Control and Information Security Policy

Purpose: To prevent unauthorized access, disclosure, modification or loss of MediLink data.

The policy requires:

1. Access to be granted according to approved job responsibilities and the principle of least privilege.
2. Administrative privileges to be restricted to authorized personnel.
3. User accounts and permissions to be reviewed periodically and when responsibilities change.
4. Sensitive identifiers and clinical information to receive appropriate protection during storage, transmission and processing.
5. Sensitive fields to be masked or excluded from reports when the intended user does not require identifiable information.
6. Audit logging to capture appropriate security and data-management events.
7. Backups to be protected against unauthorized access and periodically tested for recoverability.
8. Security incidents and suspected unauthorized disclosures to be recorded and escalated.
9. Development and testing activities to use synthetic or appropriately de-identified data wherever practicable.

National identification information must not remain exposed in plaintext throughout the trusted master-data pipeline. The design should use an appropriate combination of encryption, access restriction and protected matching mechanisms.

Where deterministic matching is required, a keyed cryptographic mechanism may be considered instead of relying solely on an ordinary hash of a predictable identifier. The key must be protected separately from the data, and the chosen mechanism must be tested for suitability.

Control evidence: Database permission tests, access reviews, configuration records, audit-log samples, backup restoration tests and security test results.

3.4 Data Retention and Disposal Policy

Purpose: To ensure that information is retained only for an authorized period and disposed of securely when retention is no longer justified.

The policy requires a documented retention schedule for patient records, clinical records, billing records, audit logs, staging data, backups and analytical datasets.

The schedule should specify:

- The data category.
- The business or legal purpose for retaining it.
- The approved retention period.
- The person responsible for authorizing retention and disposal.
- The applicable archival arrangements.
- The disposal or de-identification method.
- The evidence required to demonstrate completion.

For this project, the proposed lifecycle includes retaining clinical records for ten years and archiving them for a further seven years. These periods should be treated as **project design assumptions pending confirmation against applicable legal, clinical, contractual and institutional requirements**, not as universal statutory periods.

At the end of an approved retention period, records should be deleted, destroyed or appropriately de-identified, unless a lawful reason requires continued retention. Disposal procedures should address relevant copies, exports and backups.

Uganda's Data Protection and Privacy Act addresses retention limitation and the destruction, deletion or de-identification of personal data when the applicable retention period expires. (ULII)

Control evidence: Approved retention schedules, archival records, disposal logs and documented exceptions to scheduled disposal.

3.5 Metadata and Data Lineage Policy

Purpose: To ensure that data definitions, origins, transformations and reporting uses are documented.

Each critical dataset should have a designated owner, an agreed definition, documented fields and an identified source. Transformation processes should document significant standardization, validation, matching and aggregation rules.

The framework requires:

1. Critical fields to be described in a data dictionary.
2. Business terms and calculations to be defined consistently.
3. Source-to-target mappings to be documented.
4. Data lineage to identify how source records contribute to trusted master data and analytical outputs.
5. Changes to data definitions and transformation rules to be reviewed and documented.

6. Report owners to identify the authoritative tables and calculations used by each management indicator.

For the MediLink implementation, this includes documenting how operational visits and billing transactions contribute to the enterprise data warehouse and how data-quality indicators are calculated.

Control evidence: Data dictionary, business glossary, lineage documentation, transformation specifications and approved change records.

3.6 Reporting and Analytics Policy

Purpose: To ensure that management reports are consistent, reproducible and based on validated data.

Reports must use approved definitions and documented calculations. Measures such as total visits, unique patients, facility performance and data-quality percentages should have explicit calculation rules.

Before publication, material reports should be checked for:

- Correctness of calculations.
- Consistency with authoritative SQL results.
- Appropriate handling of missing and duplicate records.
- Consistent reporting periods and facility definitions.
- Appropriate access restrictions and suppression of unnecessary patient identifiers.
- Clear disclosure of known limitations.

Financial indicators must be reconciled against the appropriate operational billing records before being presented as authoritative revenue figures.

Control evidence: SQL outputs, reconciliation results, documented KPI definitions, report-validation records and approved dashboard screenshots.

4. Compliance Gap Management

4.1 Purpose

Compliance gap management provides a structured process for identifying weaknesses, assessing their significance, assigning corrective actions and verifying that remediation has been effective.

A compliance issue should not be considered resolved merely because a policy or procedure has been written. Closure requires appropriate evidence that the corrective action has been implemented and that the relevant control works as intended.

4.2 Identified Project Gaps

The project scenario identifies the following issues requiring attention.

Gap	Status	Risk	Proposed remediation	Verification evidence
National identification information may remain exposed in plaintext within the trusted master-data pipeline.	OPEN	Unauthorized disclosure or misuse of sensitive identity information.	Apply appropriate encryption, restrict access, review staging and master-data transformations, and protect any cryptographic keys used for identity matching.	Schema and configuration review, access tests, inspection of relevant stored values and documented security test results.
A documented data-breach response process is required.	OPEN	Delayed detection, escalation, investigation and containment of privacy or security incidents.	Establish an incident response procedure defining reporting channels, responsibilities, evidence preservation, risk assessment, containment, recovery and applicable notification obligations.	Approved procedure, incident-response exercise, documented test results and corrective-action records.

The first gap should be assessed across the complete data flow, including source ingestion, staging, matching, trusted storage and analytical outputs. Protecting one database column does not establish that the information is protected throughout the pipeline.

The second gap requires more than an incident-response document. Relevant personnel should understand their responsibilities, and the process should be tested through a suitable exercise.

4.3 Compliance Register

Each identified compliance issue should be recorded in a register containing:

- Unique issue identifier.
- Description of the gap.
- Relevant requirement or control.
- Risk assessment.
- Responsible owner.
- Agreed corrective action.
- Target completion date.
- Current status.
- Verification evidence.
- Closure approval.

The status of an issue should be reported accurately as open, in progress, pending verification or closed.

4.4 Regulatory Alignment

The proposed framework should be reviewed against Uganda's Data Protection and Privacy Act, 2019, and Data Protection and Privacy Regulations, 2021. Relevant obligations should be mapped to specific policies, procedures and technical controls rather than addressed through a general statement that the project is compliant.

The review should consider applicable requirements relating to lawful processing, accountability, data quality, security safeguards, retention, data-subject rights and the handling of personal data breaches.

The framework does not, by itself, demonstrate legal compliance. A real implementation would require a documented legal and compliance assessment, confirmation of applicable obligations and evidence that the relevant controls are operating effectively.

5. Data Governance Monitoring and Performance Measurement

Governance effectiveness should be assessed through defined indicators rather than through the existence of policies alone.

The following indicators are proposed for the MediLink project.

Indicator	Definition	Proposed target
Patient telephone completeness	Percentage of patient master records with a populated telephone number.	At least 97%.
Potential duplicate monitoring	Number of potential duplicate records identified and awaiting review.	Monitor trend and ensure all identified cases are assigned for review.
Data-quality issue resolution	Percentage of recorded data-quality issues resolved or formally escalated within their agreed service period.	Target to be approved by the governance lead.
Referential integrity	Percentage of tested relationships that satisfy the defined integrity requirements.	100% for applicable enforced relationships.
Access review completion	Percentage of scheduled access reviews completed and documented.	100% of scheduled reviews.
Report validation	Percentage of designated critical reports that pass documented validation before publication.	100% of designated critical reports.

Monthly reporting timeliness	Percentage of required monthly reports delivered by the approved deadline.	100% by the fifth day of the following month, where this is the agreed project requirement.
Compliance remediation	Percentage of compliance issues closed with documented verification evidence.	100% of issues marked as closed must have supporting evidence.

These are proposed governance targets, not measured results. Actual performance should be calculated from the database, issue register, access-review records and reporting evidence.

5.1 Monitoring Responsibilities

The data steward should monitor data-quality results and unresolved exceptions. IT custodians should provide relevant technical evidence, including access-control and system-monitoring records. Report owners should validate reporting outputs, while the Data Governance Lead should review significant issues and coordinate corrective actions.

Monitoring results should be documented at agreed intervals. Material deviations from targets should trigger investigation, assignment of corrective actions and follow-up verification.

6. Governance Review and Continuous Improvement

The framework should be reviewed periodically and whenever a significant change affects data processing, system architecture, access arrangements or applicable requirements.

Review triggers include:

- Introduction of a new source system or healthcare facility.
- Changes to patient-matching rules or master data definitions.
- Introduction of new categories of personal or clinical information.
- Changes to data-retention requirements.
- Significant security incidents or audit findings.
- Repeated data-quality failures.
- Changes to reporting definitions or warehouse transformations.
- Changes in applicable legislation or regulatory guidance.

The review should assess data-quality trends, unresolved governance issues, access-control results, audit findings, incident records, report-validation outcomes and the status of corrective actions.

Changes to the framework should be documented, reviewed by the appropriate owner and approved by the designated authority. Previous versions should be retained where necessary to provide an audit trail.

A review should result in documented decisions, assigned actions and target dates. Where corrective actions are required, their effectiveness should be verified before closure.

7. Implementation Approach

The framework should be introduced in stages to make responsibilities and controls manageable.

Stage	Key activities	Expected output
1. Governance establishment	Assign data owners, stewards and technical custodians. Approve governance responsibilities.	Governance responsibility register.
2. Standards and policies	Approve data definitions, identity rules, quality requirements, access rules and retention schedules.	Approved policy and standards documents.
3. Technical implementation	Apply approved database constraints, validation rules, access controls, logging and data-protection mechanisms.	Implemented controls and technical evidence.
4. Monitoring and validation	Run data-quality checks, validate reports, review access and test selected controls.	Scorecards, test results and monitoring records.
5. Review and improvement	Evaluate results, resolve gaps and revise policies where necessary.	Governance review records and corrective-action register.

For the capstone, the implementation evidence should distinguish completed technical work from proposed organizational procedures. For example, a database constraint can be demonstrated through a schema inspection or test, while the appointment of a Data Governance Lead requires organizational evidence if it is claimed to have occurred.

8. Conclusion

The MediLink Health Network Enterprise Data Governance Framework establishes a proposed structure for managing data ownership, quality, patient identity, information security, compliance and reporting throughout the enterprise data lifecycle.

By combining defined responsibilities, documented policies, technical controls, measurable indicators and evidence-based review, the framework provides a basis for improving the reliability and appropriate use of integrated healthcare data.

Its effectiveness depends on more than the existence of written policies. Responsibilities must be assigned, controls must be implemented, exceptions must be investigated and performance must be measured. Compliance claims must be supported by appropriate evidence, and retention,

privacy and security arrangements must be validated against the requirements applicable to the real operating environment.

9. References

1. Republic of Uganda. (2019). *Data Protection and Privacy Act, 2019*. Chapter 97. Official consolidated Act via Uganda Legal Information Institute
2. Republic of Uganda. (2021). *Data Protection and Privacy Regulations, 2021*. Statutory Instrument No. 21 of 2021. Official regulations via the National Information Technology Authority-Uganda